

ColliderAgent revision: audit of open placeholders measurements of 2026-09-16/17 and 2026-09-24, and proposed text

2026-09-24

Legend

DONE = measured today, text below can be pasted. **PARTIAL** = partly measured; the rest is in progress. **RUNS** = needs the remaining benchmark runs (Opus 4.6 $\times 3$ per benchmark; cross-model). **AUTHOR** = a decision or confirmation only the authors can make.

1 Table S2 (sm.pdf p.4) — software environment

Status: **DONE** for every package (all read inside the images); only the Claude Code version of the *original* runs is open (see note). All versions were read inside the images on zhustation on 2026-09-16 with a probe blueprint (**collider-env-probe**; job logs are cited per row). Two corrections to the current table: the analysis Python stack lives in the *agent runtime*, not in the collider image (the image has no **uproot**), and Delphes is 3.5.1.

Image / runtime	Package	Version
collider	MadGraph5_aMC@NLO	3.7.0
collider	Pythia8	8.316
collider	Delphes	3.5.1
collider	MadAnalysis5	1.11.0 (2025-04-23)
collider	LHAPDF	6.5.5
collider	HepMC	2.06.09
collider	ROOT	6.28/10
collider	Python	3.10.12; numpy 1.26.4, scipy 1.15.3, awkward 2.9.0, matplotlib 3.10.8
mma-het	Wolfram Engine	13.3.0
mma-het	FeynRules	2.3.49 (29 Sep 2021)
micromegas	micrOmegas / CalcHEP	6.3.0 / 3.9.2
agent runtime	Python analysis stack	Python 3.13.12; uproot 5.7.4, awkward 2.9.0, numpy 2.4.4, scipy 1.15.3
agent runtime	Claude Code	2.1.273 (re-runs of 2026-09-16); original runs: see note
agent runtime	LLM	Claude Opus 4.6 (claude-opus-4-6), still served on 2026-09-16

Proposed replacement rows (booktabs, same layout as the current table):

```
collider    & MadGraph5_aMC@NLO & 3.7.0 \\
            & Pythia8      & 8.316 \\
            & Delphes    & 3.5.1 \\
            & MadAnalysis5 & 1.11.0 \\
            & LHAPDF     & 6.5.5 \\
```

	& HepMC	& 2.06.09 \\
	& ROOT	& 6.28/10 \\
	& pyhf	& 0.7.6 \\
mma-het	& Wolfram Engine	& 13.3.0 \\
	& FeynRules	& 2.3.49 \\
agent runtime	& Claude Code	& 2.1.x (AUTHORS: version of the original runs from the usage logs the 2026-09-16 re-runs used 2.1.273) \\
	& LLM	& Claude Opus 4.6 \\
	& uproot, awkward, numpy, scipy	& 5.7.4, 2.9.0, 2.4.4, 1.15.3 \\

Note on the Claude Code version: no session log from the April–May runs exists on this machine (the earliest local logs are from August 2026), so the original version has to come from the co-author who ran them (**AUTHOR**). The rows above are otherwise complete.

2 sm.pdf p.5, “LLM inference accounts for a minor fraction [AUTHORS: fill in numbers ...]”

Status: **PARTIAL** — measured for the ALP EFT benchmark with three current models; the Opus 4.6 numbers for all eight benchmarks are being produced now. The claim as written is *not* supported for the light benchmarks: attributing every gap before a model message to inference and every gap before a tool result to tool execution (main session plus stage subagents), the model share of wall-clock was 38% (Opus 5), 54% (Opus 4.8) and 51% (Sonnet 5); time inside `magnus run/status` calls was 60%, 45% and 47% (the two overlap when a stage polls a running job). Proposed wording (with the 21 Opus 4.6 runs of Table S3, model share 0.23–0.80, median 0.55):

For the light benchmarks of Table S3 the wall-clock time splits roughly evenly between LLM inference and cluster jobs (model share 0.2–0.8, median 0.55); for the Dark-SMEFT campaigns, where each campaign submits more than 100 jobs, cluster time dominates.

3 Table S3 (sm.pdf p. 6) — resource usage; also paper.pdf p. 5 and p. 7, reply.pdf p. 4 and p. 6

Status: **PARTIAL** — Claude Opus 4.6 (effort `xhigh`, v2 skills, cold memory), three attempts per benchmark for the seven figure benchmarks (attempt 1 launched 2026-09-16 22:00 UTC, attempts 2–3 on 2026-09-24). The two Delphes-level benchmarks had one attempt each at the time of this audit (footnotes a, b); their row shows that single attempt. **Update 2026-09-28:** attempts 2–3 succeeded on both (Opus 4.6 2/3 each, attempt 1 now counted as an infrastructure failure) and Opus 5 completed both at the first attempt; see `experiment-report-2026-09-26.pdf`, Part I. Values are means over the successful attempts ($n = \text{successful/attempted}$; per-attempt values in `docs/paper/results-2026-09-24/runs.csv`, figures in `.../figures/`). Tokens = main session + all sub-agents (uncached + cache-write + cache-read); files = distinct paths written or edited; Magnus jobs = `magnus run` submissions; USD from the CLI’s `total_cost_usd`.

Benchmark	n	Wall [h]	Sub-agents	Magnus jobs	Files	Tokens in [M]	Tokens out [k]	USD	Result
Scalar LQ, m_{ej} [Fig. 2(a)]	0/1	6.41	2	42	28	214.0	220	–	TBD ^a
ALP EFT, $\not{E}_T$ [Fig. 2(b)]	3/3	1.01	3.0	6.7	16.7	15.1	31	12.8	success*
U(1)' scan [Fig. 2(c)]	3/3	0.92	3.3	17.0	22.3	4.4	30	7.6	success*
U1 LQ mono- τ [Fig. 2(d)]	0/1	4.00	2	29	19	181.4	219	106	TBD ^b
Heavy N [Fig. S2(a,b)]	3/3	0.60	3.0	9.0	14.3	3.9	17	4.8	success*
General Z' [Fig. S2(c)]	3/3	0.73	3.0	7.3	14.3	8.3	25	7.2	success*
KK graviton [Fig. S2(d)]	3/3	0.95	2.7	9.3	14.3	3.8	28	6.4	success*
U1 LQ at MuC, η [Fig. S2(e)]	2/3	0.68	3.0	14.0	6.0	4.4	18	7.8	success*, 1 failure ^c
U1 LQ at MuC, reach [Fig. S2(f)]	3/3	1.22	3.0	19.0	21.7	13.3	39	12.7	success*

* end-to-end reproduction with a quantitative deviation recorded as a footnote in at least one attempt (texts in `docs/paper/results-2026-09-24/tables.md`). Ranges over the 21 attempts of the seven figure benchmarks: 0.37–1.41 h, 3.6–22.2 USD, 1.3–36.9 M input tokens.

^a **TBD (handed over)**. Run stopped by the operator after 6.4 h without a figure. Model, validation and the 100 000-event LUXlep parton sample (lepton→photon LHE workaround) were done; two Pythia8+Delphes launches of the full sample finished on the cluster but failed at upload with “No space left on device” (the Delphes ROOT file alone is 27 GB against the job’s 10 GB packing space); a third launch that deletes HepMC/ROOT after the run (MG5 ! shell lines after the final `done`, keeping the LHCO) was running when stopped. The v2 skills now document the limit and the cleanup (commit 61800a9); a rerun is expected to take 1.5–2 h.

^b **TBD (handed over)**. Attempt 1 generated all 18 Delphes runs (4.0 h, 106 USD; four 9-run launches were lost to the same disk limit before the agent split them), then the orchestrator ended its turn with “waiting for event file downloads” and the headless session terminated without the analysis stage (failure mode (iv) in the paper’s classification). A continuation session in the same workspace produced the figure in 1.7 h / 29.5 USD with the reference’s shape but a 20–35 % weaker exclusion. The orchestrator skill now forbids ending a turn to wait (commit e7c5b61).

^c Attempt 2 of U1 LQ at MuC, η : both leptoquark curves coincide with the SM background to 2–3 % per bin, i.e. the new-physics contribution did not enter event generation (failure mode (i), *model*); attempts 1 and 3 reproduce the reference (attempt 1 without triggering the orchestrator: the main session ran the pipeline itself, 0 sub-agent calls).

Model-latency share of wall-clock (transcript timestamps) over the 21 attempts of the figure benchmarks: 0.23–0.80, median 0.55; the sentence “LLM inference accounts for a minor fraction” (sm.pdf p. 5) should be replaced by, e.g.:

For the figure benchmarks of Table S3 the wall-clock time is shared about equally between LLM inference and cluster jobs (model share 0.2–0.8, median 0.55 over 21 runs); cluster time dominates only for the Delphes-level benchmarks and the Dark-SMEFT campaigns.

4 Tables S4 and S5 — state on 2026-09-24

Table S5 (robustness, Opus 4.6 $\times 3$) is complete for the seven figure benchmarks: 20 of 21 attempts succeeded (the one failure is footnote *c* above). The two Delphes-level benchmarks have one attempt each (TBD^a, failure^b) and still need their $\times 3$ (1.5–6 h and up to >100 USD each at **xhigh**; use `--effort high`).

Table S4: column A = Opus 4.6 with Claude Code; column B = GPT-5.5 with Codex CLI 0.155.1 on branch `codex-com` (skills-only mode, same prompts, one clean directory per attempt, same judge, metrics from the Codex event stream and the main/sub-agent thread rollouts). GPT-5.3-Codex, the model contemporaneous with Opus 4.6, is not served to a ChatGPT-login Codex account (an API key is required), so GPT-5.5 was used on 2026-09-24; on 2026-09-26 column B was re-run with GPT-5.3-Codex through an API key and column C (Gemini via ADK) was measured — see the next section.

Benchmark	Table S4 columns			single runs, 2026-09-16, default effort		
	A: Opus 4.6 ($\times 3$)	B: GPT-5.5 / Codex	C	Opus 4.8	Opus 5	Sonnet 5
Scalar LQ	0/1 ^a	1/1 ^{*e} (4.41 h, 24 jobs, 75.0 M tok)	–	–	–	–
ALP EFT	3/3	1/1 (0.99 h, 9 jobs, 13.4 M tok)	–	1/1	1/1	1/1
U(1) ['] scan	3/3	1/1 (1.03 h, 21 jobs, 11.1 M tok)	–	–	1/1 [*]	–
U1 LQ mono- τ	0/1 ^b	1/1 ^{*d} (2.55 h, 20 jobs, 49.4 M tok)	–	–	–	–
Heavy N	3/3	–	–	–	–	–
General Z [']	3/3	–	–	–	–	–
KK graviton	3/3	–	–	–	–	–
U1 LQ at MuC, η	2/3 ^c	–	–	–	–	–
U1 LQ at MuC, reach	3/3	–	–	–	–	–

^d Codex/GPT-5.5 completed mono- τ end to end (18 Delphes samples, 180 000 events): the $R_{D^{(*)}}$ bands match the reference, the exclusion boundary has the reference’s shape but is about 30 % weaker ($\sqrt{|g_c g_b|} \approx 1.07$ at 0.8 TeV against ≈ 0.8). The Opus 4.6 continuation run of attempt 1 was 20–35 % weaker in the same way, so two independent implementations agree with each other and not with the paper; the deviation is more likely a property of the analysis procedure the prompt prescribes (the LHCO output carries no separate τ collection; both runs used type-4 jets as hadronic- τ proxies) than an agent error.

^e Codex/GPT-5.5 completed Scalar LQ end to end in 4.4 h: 100 000 LUXlep parton events, lepton $\rightarrow$ photon LHE substitution, Pythia8+Delphes (ATLAS card) to LHCO after two failed detector passes (the interactive re-shower could not materialise `pythia8_card.dat`; “No space left on device”), the third as a non-interactive launch. The m_{ej} signal peaks at 3 TeV with ≈ 1.5 events/bin/100 fb^{−1} against ≈ 3 in the reference (factor 2 low) and ends at 3.8 TeV; the Delphes card used anti- k_T $R = 0.6$ instead of 0.4. Opus 4.6 attempt 1 was stopped at the same stage after 6.4 h, so the benchmark is completable with the current skills and its Opus 4.6 $\times 3$ is worth running.

Correction to the U(1)['] note of 2026-09-17: the Codex/GPT-5.5 run reproduces the paper’s 3 TeV contour tip ($\tilde{g} \approx -0.72$, $g_1' \approx 0.65$), whereas all three Opus 4.6 attempts and the Opus 5 run give $g_1' \approx 0.48$. The 25 % shortfall is therefore an analysis choice made in the run, not a limitation of the prompt; the earlier wording “systematic to the prompt’s analysis procedure” is withdrawn (it stays a footnote, the runs stay successful under the agreed criterion). The 2026-09-26 batch (next section) shows the same 0.48 tip in GPT-5.3-Codex and GPT-5.5/ADK runs, so it is not Claude-specific either.

Success criterion used (agreed 2026-09-16): the run completes end to end and the figure agrees with the requested part of the reference qualitatively; quantitative deviations go to a footnote (*). Failure modes follow the paper’s four classes; every verdict was produced by the assistive judge (`scripts/bench/judge.sh`) and confirmed by eye.

Still open after 2026-09-26:

1. Scalar LQ and mono- τ , Opus 4.6 $\times 3$: `scripts/bench/run_benchmark.sh` 2005.06475 2 `claude-opus-4-6 --effort high` and ... 1811.07920 3 ...; both root causes are fixed in the skills.
2. $\times 3$ for the Opus 5 and GPT-5.3-Codex columns (one attempt each so far; about 1 h per run).
3. A stricter model/harness separation: an Anthropic model inside the ADK loop (needs an Anthropic API key through LiteLLM).

5 Model C and the comprehensive Table S4 (2026-09-26)

Model C is the repository’s own `python-agent`: one Google ADK `LlmAgent` (ADK 2.10, LiteLLM) with 13 flat tools (read/write/edit, FeynRules validation, UFO generation, MadGraph compile/launch, MadAnalysis, YAML workflow, plus `run_python/run_shell` added on 2026-09-26 so that a figure can be drawn at all), no sub-agents, no skills, no memory, no context compaction, `max_llm_calls=150` (300 for the two Delphes benchmarks), a 6 h wall limit. Model: `gemini-3.1-pro-preview` (released 2026-02-19, the Gemini contemporaneous with Opus 4.6) through an OpenAI-compatible relay. Two controls

separate the harness from the model: the same Gemini model inside Gemini CLI 0.61 (a coding agent, same `.agents/skills` as the Codex harness, no custom sub-agents) and GPT-5.5 inside the same ADK loop. Column B was re-run with GPT-5.3-Codex (API key). Same prompts, one clean directory per attempt, same judge; every run was checked with `scripts/bench/job_provenance.py` (footnote *f*).

Benchmark	A: Opus 4.6 Claude Code ($\times 3$)	A': Opus 5 Claude Code	B: GPT-5.3-Codex Codex CLI	C: Gemini 3.1 Pro ADK ($\times 3$)	C ₁ : Gemini 3.1 Pro Gemini CLI	C ₂ : GPT-5.5 ADK
Heavy N	3/3	1/1	1/1	1/3	1/1	1/1
U(1)' scan	3/3	1/1	1/1	1/3	1/1	1/1
ALP EFT	3/3	1/1	1/1	2/3	1/1	1/1
General Z'	3/3	1/1	1/1	1/3	1/1	1/1
KK graviton	3/3	1/1	1/1	2/3	1/1	1/1
U1 LQ at MuC, η	2/3	1/1	0/1	0/3	1/1	1/1
U1 LQ at MuC, reach	3/3	running	0/1	1/3	1/1	1/1
figure benchmarks	20/21	6/6	5/7	8/21	7/7	7/7
Scalar LQ (Delphes)	0/1 ^a	–	0/1 ^g	0/1 ^j	–	0/1 ^h
U1 LQ mono- τ (Delphes)	0/1 ^b	–	1/1 [*]	0/1 ⁱ	–	0/1 ^h

Long-horizon indicators, figure benchmarks, means per run (the ADK event log records every model call and every tool's success flag; the other harnesses do not expose comparable counts):

Column	runs	ok	wall [h]	LLM calls	tool calls	tool errors	refine cycles	max ctx [k tok]	exit
Gemini 3.1 Pro / ADK	21	8	1.41	117	117	39	39	120	10 capped
GPT-5.5 / ADK	7	7	0.71	35	39	4	4	71	completed
Gemini 3.1 Pro / Gemini CLI	7	7	0.63	–	72	–	–	–	completed
GPT-5.3-Codex / Codex	7	5	0.81	–	–	–	–	–	–
Opus 5 / Claude Code	4	4	1.22	–	–	–	–	–	–

Reading: (i) the same Gemini model goes from 8/21 in the single ADK loop to 7/7 in a coding-agent harness, with half the wall-clock and no budget exhaustion — the harness, not the model, is the main factor; (ii) GPT-5.5 survives the same loop (7/7, 35 calls, 4 errors per run), i.e. the strongest current model masks the structural penalty on the figure benchmarks; (iii) the Delphes-level benchmarks separate them again: the coding agents solve the lepton $\rightarrow$ photon LHE rewrite and the 10 GB job limit by writing scripts and re-launching (Codex GPT-5.5 4.4 h, GPT-5.3-Codex 2.6 h for mono- τ with the paper's exclusion), whereas GPT-5.5 in the ADK loop stopped at the parton level and smeared m_{ej} by hand (footnote *h*); (iv) ADK failures are loops — 10 of 13 are the 150-call budget spent in event generation or model building (108/78/121 failed tool calls and 379k/182k/230k-token contexts in the three MuC η attempts) — while the Codex/Claude failures are single physics errors (the MuC η coupling not taking effect, an empty contour plot).

^f The Magnus job list is shared by all runs. The first GPT-5.3-Codex Heavy N run “reproduced” the figure in 7 minutes from the logs of the 2026-09-16 Opus 4.6 jobs; it was quarantined and re-run, every harness prompt now carries `scripts/bench/benchmark_rules.md` (submit your own jobs), and `job_provenance.py` verified all other 52 runs of this batch. Earlier batches were not scanned (sandboxes deleted) but built their own models (5–42 jobs each).

^g Model and UFO done; the agent could not install the LUXlep PDF set (`--pdf` installer lacks `lhapdf-config` on zhustation) and stopped after 0.8 h instead of using the MG5-internal `pdlabel/lhaid` route that the Opus 4.6 and GPT-5.5 runs used.

^h 100k LUXlep parton events, no Pythia8/Delphes/selection: the agent reported that the tool interface “did not support the requested LHE lepton-to-photon replacement before showering” and applied a detector-resolution approximation to the parton-level m_{ej} (peak ≈ 180 events/bin against ≈ 4 in the reference); counted as a generation failure. The same agent's mono- τ run attempted three Pythia8+Delphes launches, reported that the LHCO output “could not be downloaded” and computed the ATLAS/CMS selections from LHE-level kinematics (boundary 0.55 at 0.8 TeV, far stronger than the paper's 0.8); also

a generation failure.

ⁱ Cap raised to 300 calls: model and UFO done, MadGraph compile failed 10 times, 277 of 300 tool calls were `run_shell` re-inspections of the same files; no events.

^j Cap 300: model, UFO and compile done, then 13 of 14 launches failed (81 error-refine cycles, context 199k tokens, 2.8 h); no sample analysed. Delphes benchmarks overall: coding agents 4/6, ADK 0/4.

U(1)' 3 TeV tip, updated: $g'_1 \approx 0.48$ in Opus 4.6 $\times 3$, Opus 5, GPT-5.3-Codex and GPT-5.5/ADK; 0.55 in Gemini CLI; the paper's 0.65 only in Codex GPT-5.5 — a common analysis choice, not a property of one model or of the prompt (this supersedes the 2026-09-24 wording). Likewise for mono- τ : the Claude continuation and Codex GPT-5.5 are 20–35 % weak, GPT-5.3-Codex reproduces the paper (0.88 vs 0.8 at 0.8 TeV).

Data: docs/paper/results-2026-09-26/ (README.md in Chinese, tables.md, runs.csv, longhorizon_*.md, figures/); harnesses scripts/bench/run_benchmark_adk.sh and run_benchmark_gemini.sh.

6 Other markers

Where	Marker	Status	Proposed action
paper.pdf p. 5	“skills-only mode executes the hello-world workflow of Sec. S-III [confirm that this has been tested]” (Codex/Cursor)	AUTHOR	The Codex-compatible layout exists (branch <code>codex-com: .agents/skills, .codex/agents, AGENTS.md</code>). Run the hello-world prompt with Codex on that branch once and keep the sentence; otherwise weaken to “is designed to”.
paper.pdf p. 7, Table I caption	“tokens consumed and wall-clock time [fill in]”	RUNS	Same numbers as Table S3 (Opus 4.6 column); cite Table S3 instead of repeating them.
paper.pdf p. 9–10	“[confirm the extent of human intervention between campaigns]”	AUTHOR	Dark-SMEFT campaigns were not re-run; statement must come from the campaign logs.
paper.pdf p. 10; reply.pdf p. 6	“several weeks of work for an experienced phenomenologist [adjust estimate]”	AUTHOR	Judgement call; unchanged by measurements.
reply.pdf p. 1	checklist (1)–(4): fill S3/S4/S5, confirm human intervention, set <code>\markupfalse</code> , remove red markers	PARTIAL	(1) in progress, (2) author, (3)–(4) mechanical once (1)–(2) are done.
reply.pdf p. 4	“the LLM token consumption [fill in from the usage logs]”	RUNS	From Table S3; note the main+sub-agent definition.
reply.pdf p. 4	“The wall-clock time of a benchmark (1–6 hours) ...”	DONE	Consistent with today's runs (0.9–1.4 h for the lightest benchmark); keep.
sm.pdf p. 4	“archived release ... DOI and per-run provenance packaged as Workflow Run RO-Crates [confirm or remove]”	AUTHOR	No RO-Crate export exists in the repository today; either remove or keep as a plan.
sm.pdf p. 5	Table S4 placeholder text “e.g. Claude Opus 4.6 vs. Claude Sonnet, GPT-5.x via Codex, Gemini via ADK”	PARTIAL	Replace with the three columns chosen above; drop Gemini/ADK unless the python-agent path is run.

Where	Marker	Status	Proposed action
sm.pdf p. 6	Table S3 Dark-SMEFT rows (sub-agent calls, tokens)	AUTHOR	Not reproducible here; from the campaign logs.

7 Provenance of the numbers

Branch `feat/skill-evolution-v2` of the ColliderAgent repository. All benchmark data are in `docs/paper/results-tables.md` (Tables S3–S5 and every footnote text, generated by `scripts/bench/aggregate.py`), `runs.csv` (one row per attempt), `documented_runs.json` (the 2026-09-16 attempts, whose sandboxes were deleted; numbers copied from `docs/paper-tbd-audit-2026-09-16.md`) and `figures/` (the produced figure of every attempt). The sandboxes under `bench_runs/` were deleted after aggregation. Image versions come from the probe blueprint `collider-env-probe` on zhustation. Skills used are the v2 skills (context-lean rewrite of 2026-09-16), so the Opus 4.6 numbers measure the current system.